Advanced Data Viewer — User Guide

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中文版：用户操作指南_v1.10.13.pdf

Open another file while the program is running

Use the file browser or File Name field. The viewer keeps the graph layout, colours, independent Y-axis settings, and time offsets for channels that match the new file. It hides missing channels and names them in the status bar. If no plotted channel remains, it shows the new file's default channel. The zoom view and A/B cursors reset for the new file.

Shift one channel in time (ms)

Open the A/B range-measurement panel. In the **Time offset** row, choose a plotted channel, enter a signed value in milliseconds, and click **Apply**. Positive values move its curve right; negative values move it left. **Reset to zero** restores its position. Each plotted channel has its own setting.

Display only shifts the plotted curve while A/B statistics and phase difference continue to use raw timestamps. **Display and calculate** also selects that channel's A/B samples by shifted time and adjusts phase difference by its shift relative to the reference channel at the measured frequency. Source files and sample values are never changed. PNG exports follow the displayed shift; CSV retains the original time values. The feature requires a time X-axis whose unit is seconds (`s`). The API exposes `set_time_offset(channel, milliseconds, mode)`.

1. Start the viewer and open a file

1. Double-click `Advanced Data Viewer_v1.10.13.exe`. Keep its console window open while you use the viewer; closing it stops the local service. Released builds carry the version in the filename; the delivery folder holds exactly one exe and keeps older ones in `_backup\`. It is still a single self-contained file with nothing to install.
2. Click `...` next to **File Name**, select a file, then click **Open** or double-click the file. You can also paste a full path into **File Name** and click **Draw Graph**. Common text formats include `.csv` and `.tsv`.
3. The channel chips appear above the plot. Wait for **Loading...** to disappear before working with a large file.
4. Click **EN** in the top-right corner to switch to English. The button then reads **中文** and switches back to Chinese.

The window has a channel list at the top, a waveform plot in the middle, an optional range-measurement panel on the right, and a status bar at the bottom. Channel names and units come from the source file and are therefore not translated.

2. Add, hide, and remove channels

Quick method: channel chips above the plot

- Click an unhighlighted channel chip to add it to **graph 1**. Its background takes the curve colour.
- Click the highlighted chip again to hide that channel. This removes it from the plotted layout; it does not delete the source channel or change the source file. If the same channel was plotted in more than one graph, this quick toggle removes it from the plotted layout.
- In a layout with multiple graphs, the chip is still a shortcut for graph 1. Use **Graph Definition** to add a channel to another graph.

Precise method: Graph Definition

1. Click **Graph Definition**. **Number of Graphs** controls how many graphs are stacked vertically. Each **Y-Axis • Graph N** section configures one graph.
2. Coloured tags in a graph section are curves **already included** in that graph. Click a tag's `×` to remove that curve from this graph only. Click the small colour swatch to change its colour.
3. The **Channel** drop-down selects the **next channel to add**. After selecting it, click `+`. The new coloured tag confirms that it has been added.
4. Click **OK** to apply the layout. **Cancel** or the dialog's `×` closes it without applying pending changes.

Goal	What to set	Result
Several channels in one graph	Set **Number of Graphs** to `1`; select each channel in **Y-Axis • Graph 1** and click `+`	Curves share one plotting area
Separate graphs	Set **Number of Graphs** to `2`, `3`, etc.; add the desired channel in each graph section	Graphs are stacked and share the X axis
Multiple graphs with overlays	Increase the graph count, then add several channels within a selected graph section	Each graph can have its own group of curves

There can be up to 7 graphs, subject to the number of usable channels in the file. Reducing the graph count removes the extra graphs' **display settings**, never their source channels.

Independent Y scales in one graph

Add at least two curves to the same graph, then select **Independent Y** in that graph section. Each curve receives its own scale and a colour-matched axis. With only one curve, the control is disabled. New graphs retain the independent-scale setting, which takes effect once a second curve is added.

With exactly two curves and **Independent Y** enabled, the **双 Y 轴错位** slider (dual Y-axis display offset, 0–50%) separates the curves visually. It moves only the displayed curves and corresponding scales. Hover values, A–B measurements, original data, and exported CSV numbers stay unchanged.

3. Zoom, pan, and read values

- Drag left-to-right over the plot to zoom into a range. Drag right-to-left to zoom out.
- Use the mouse wheel to zoom around the pointer. Hold the right button and drag to pan. A right-click **without moving** opens the context menu.
- Double-click the plot, press `Esc` when no dialog is open, or click **Global View** to show the entire range.
- Hover over the plot to read the X position and the original values of each displayed curve. Multiple graphs, independent scales, and display offsets do not alter these values.

4. Set A/B cursors and calculate a range

1. Right-click the plot at the first time point and choose the cursor-A command (**设为光标 A**); repeat at another point for cursor B (**设为光标 B**). A right-drag pans the plot, so release the button without moving when opening the menu.
2. Hover near a vertical cursor line until the pointer changes to a horizontal resize arrow. Drag the line with the left button to reposition it. The context menu's **清除光标** command clears the cursors.
3. The status bar shows A, B, their separation `Δ`, and `1/Δ`. **`1/Δ` is the reciprocal cursor separation, not an automatically measured waveform frequency.**
4. Right-click the plot and choose **测量 A–B 区间…** to open the right panel. You can also type exact values into its A and B fields (`B > A`) and click **Measure**. **用光标 A–B** reloads the positions of the plot cursors.
5. The table covers the **currently plotted channels** and reports minimum, maximum, mean, peak-to-peak, RMS, standard deviation, mean crossings, period, frequency, slope, and valid-data percentage. Moving a cursor updates the panel automatically; **Measure** also recalculates on demand.

The range needs enough samples for meaningful statistics. A steady or nonperiodic signal may show `—` for period and frequency because no reliable cycle was found; `—` does not mean zero. Drag the left edge of the panel to resize it, or close it with the `×` in its header.

Phase difference: Plot at least two periodic signals in the same time range and set A/B. In the panel's **Phase** section, choose the reference and comparison channels. Leave frequency blank for automatic estimation or enter a known frequency in Hz, then click **Measure phase**. A positive phase means the comparison channel leads the reference. **Details…** shows other channels and fit quality. Check fit quality before relying on a result from a short or nonsinusoidal segment.

Curve fitting: The **Curve fitting** section offers linear, polynomial trends of degrees 2–6, and a single-tone sine fit. **Fit equation** reports the function, R^2, and RMSE. **Copy equation** copies it; **Overlay fit** creates a channel for the fit in the A/B range and plots it with the source curve.

5. Create a virtual calculated channel

A **derived channel** calculates a new sampled series from existing channels. It does not modify the original file.

1. After opening a file, click the toolbar's `ƒ` button, or right-click the plot and choose **New derived channel...**.
2. Enter an **Expression**. The easy way is to click a channel name in the dialog: it inserts the correct reference and adds quotes when the name contains spaces or hyphens, e.g. `"Ch A-1" / 2`.
3. Enter a **Name** and **Unit**. A blank name becomes `derived_N`; the unit may be left blank, but when given it shows in the legend and the channel chips.
4. Click **Test** to preview minimum, maximum, mean and valid-point percentage. A test does **not** register a channel, so iterate as often as you like.
5. Click **Create and plot**. The new channel appears in the top list and is plotted in graph 1. To place it in another graph, use that graph's **Channel** drop-down and `+` in **Graph Definition**.

Available functions

Form	What it does	Watch out
`a + b`, `a * 2`, `a ** 2`, `a / b`	Point-wise arithmetic and powers; channels and numbers mix freely	Division by zero gives NaN/inf -- check **Test** first
`abs(a)`	Absolute value	
`sqrt(a)`	Square root	NaN where the input is negative
`log(a)`, `exp(a)`	Natural log / exponential	`log` needs positive input
`min(a, b)`, `max(a, b)`	Smaller / larger of two values, point-wise	Compare with a constant to clip: `max(a, 5)`
`diff(a)`	Difference between neighbouring samples	**Does not divide by time**; for a derivative use `/ diff(x-axis channel)`
`integ(a)`	Cumulative integral	Accumulates with the file's median sample interval; treat as indicative when the interval varies
`lp(a, 10)`, `hp(a, 10)`	One-way first-order low/high pass, 10 Hz cutoff	Adds -atan(f/fc) of phase; do not use it for phase work
`lp0(a, 40)`, `hp0(a, 40)`	Zero-phase low/high pass, 40 Hz cutoff	Phase preserved; use these for delta and tan-delta
`movmean(a, 51)`	Centred moving average over 51 **samples**	Zero phase, but the window counts samples, not time

Form	What it does	Watch out
`shift(a, 3)`	Shift the whole series by 3 samples	Positive shifts later, negative earlier; also in samples
`pi`, `e`	Constants	e.g. `2*pi*236`

Channel names may be fragments (case-insensitive, shortest match wins); use the full name when ambiguous. Function names must be written exactly as in the table, and the first argument inside the brackets must be a channel. Derived and result channels (marked `ƒ` or `Σ`) can be used as inputs too.

Examples you can adapt

1. **kN to stress (MPa)**: `Force * 1000 / 506.7`, unit `MPa`. 506.7 is the cross-section of a 50.8 mm diameter specimen in mm² -- change it with the specimen.
2. **Displacement to strain**: `Disp / 50.8`, no unit. 50.8 is the gauge length in mm.
3. **Remove an installation zero**: `Radial_compensated - 0.0012`, unit `mm`. Take the constant from the unloaded segment.
4. **Derivative (mm/s)**: `diff(Disp) / diff(Time)`, unit `mm/s`. Always divide by the time difference -- `diff` alone is just a sample-to-sample difference and can be off by tens of times where the interval changes.
5. **Phase compensation (cancel the 236 Hz first-order lag)**: `Disp + diff(Disp)/diff(Time)/(2*pi*236)`, unit `mm`. Equivalent to advancing the channel by about 0.674 ms; amplitude rises slightly with frequency (about +9% at 100 Hz).
6. **Zero-phase low pass, suppress the high-frequency chain resonance**: `lp0(Radial_compensated, 40)`, unit `mm`. Phase unchanged; amplitude falls to 0.62 at 31.6 Hz and 0.17 at 90 Hz.
7. **Health-check the samples**: `diff(Time)`, unit `s`. A flat stretch means a gap; a spike means a timestamp jump -- faster than reading logs.

> `Time` above is the x-axis channel of the current file; click the actual x-axis name when you change files.

Use `lp0` / `hp0` for phase and loss work, never the one-way `lp` / `hp`. With a 40 Hz cutoff, `lp` already adds -14 degrees at 10 Hz and -38 degrees at 31.6 Hz, which a loss calculation will happily read as material damping. Filtering also changes amplitude (gain `1/(1+(f/fc)^2)`), so filter both channels with the same cutoff before comparing an amplitude ratio. Each end has a few seconds of transition -- stay away from it when analysing.

The upper half of the derived-channel dialog lists existing derived channels with **Plot / Edit / Delete** actions. **Edit** restores the expression and changes the action button to **Save changes**. Renaming recreates the channel under the new name and drops the old name. **Delete** requires a second confirmation and removes the channel from all graphs. Original source channels cannot be deleted there.

6. Turn analysis results into channels

The **Range measurement**, **Cycles**, and **Segments** sections in the right panel each have a **Save as channel** button. Enter an optional name in **Result → channel → Name** at the bottom of the panel. In the relevant section, select the value to retain, such as mean, peak-to-peak, amplitude, or RMS, then click that section's **Save as channel**. A blank name is generated automatically.

Result channels have a `Σ` marker and a **result** tag. They can be plotted, used in expressions, and analysed again. A saved range statistic is a constant line within its range; per-cycle and per-segment values form steps. Result channels do not have editable expressions—delete and recompute one if needed.

For cycles, choose a channel and a split method (**Peak/valley pairs** or **Zero crossings**), then click **Run**. For segments, choose a channel and split by **time** or **cycle**; when splitting by time, set the segment length in seconds and click **Run**. These calculations use a valid A/B range when one is set.

7. Save actual values and pictures to disk

Derived and result channels live only in the current open-file session. They are not written back to the source data files. Closing the program or opening another file means you must recreate them. **Save as channel** keeps a result available for further plotting and analysis in the session. Export a CSV to retain its sampled values on disk.

Save curve data, including a virtual channel

1. Make sure the channel is **plotted**. Graph CSV export includes the X axis and currently plotted channels; an unplotted channel is omitted.
2. Click **Global View** first if you need the entire dataset. If the graph stays zoomed in, only the current view range is exported.
3. Click **Export Graph → Export CSV (current view)**. The program creates a CSV in the `exports` folder next to the EXE and shows its full path and row count. **Export Graph → Open export folder** helps locate it.
4. The CSV contains original channel values. Independent Y scales and dual Y offsets do not change the exported numbers.

Export Graph → Export PNG (current view) saves the current plot to the same `exports` folder. **Copy image to clipboard** is useful when preparing a report.

The **Export CSV** buttons in the right measurement panel serve a different purpose: range measurement exports a **statistics table**, while the segment section exports **per-segment statistics**. The browser saves those downloads to its configured download folder. They are not full point-by-point curve exports.

8. Common questions

Symptom	What to do
I selected a channel in Graph Definition, but no curve appeared	Click that graph's `+`, confirm that a coloured tag appears, then click **OK**.
Independent Y is disabled	Add a second curve to the **same graph**.
The dual Y-axis offset cannot be moved	The graph must have exactly two curves and **Independent Y** must be enabled.
Clicking a top channel chip adds it to graph 1	Use **Graph Definition** to add it to the intended graph section.
The A/B result covers the wrong interval	Check both cursor lines or type exact A and B values in the right panel, then click **Measure**. B must exceed A.
The CSV lacks my virtual channel or only has part of the data	Plot the virtual channel before export; click **Global View** for the full range.
My virtual channel vanished after reopening	It was session-only. Export its values as CSV, and recreate its expression in a new session.

The API manifest is available at `http://127.0.0.1:8770/api/tools` (use the actual port if changed). This guide focuses on GUI operation. For questions or feedback, contact **bob.zhang19s@gmail.com**.